ASP.NET MVC with AJAX and REST APIs

What is AJAX?

AJAX (Asynchronous JavaScript and XML) is a technique that allows web pages to be updated asynchronously by exchanging small amounts of data with the server behind the scenes. This means you can update parts of a web page without reloading the whole page.

Benefits of using AJAX:

- Faster and smoother user experiences
- Reduces bandwidth usage
- Enables partial page updates

How is it used? AJAX typically involves:

- Making an HTTP request using JavaScript (often using jQuery's `$.ajax()` function)
- Sending or receiving JSON data from/to the server
- Dynamically updating the HTML DOM using JavaScript/jQuery

Entity and ViewModel:

```
public class student
{
   public Guid Id { get; set; }
   public string Name { get; set; }
   public string Email { get; set; }
   public string Phone { get; set; }
   public bool Subscribed { get; set; }
}
```

AddStudentViewModel:

```
public class AddStudentViewModel
{
   public string Name { get; set; }
   public string Email { get; set; }
   public string Phone { get; set; }
   public bool Subscribed { get; set; }
}
```

ApplicationDbContext

```
public class ApplicationDbContext : IdentityDbContext
{
    public ApplicationDbContext(DbContextOptions<ApplicationDbContext> options) :
base(options) { }

    public DbSet<student> students { get; set; }
}
```

StudentController with API Endpoints

```
[HttpGet]
public async Task<IActionResult> GetStudentsJson()
{
    var students = await dbContext.students.ToListAsync();
    return Json(students);
}

[HttpPost]
public async Task<IActionResult> AddStudentAjax([FromBody] AddStudentViewModel
viewModel)
{
    var student = new student
    {
        Id = Guid.NewGuid(),
        Name = viewModel.Name,
        Email = viewModel.Email,
        Phone = viewModel.Phone,
        Subscribed = viewModel.Subscribed
    };

    await dbContext.students.AddAsync(student);
    await dbContext.SaveChangesAsync();

    return Json(new { success = true, message = "Student added successfully" });
}
```

List View with AJAX (List.cshtml)

```
@{
    ViewData["Title"] = "Student List";
}
```

```
<h2>Student List (AJAX)</h2>

<table class="table" id="studentTable">
    <thead>
        <tr>
            <th>Name</th>
            <th>Email</th>
            <th>Phone</th>
            <th>Subscribed</th>
        </tr>
    </thead>
    <tbody>
        <!-- Will be filled by AJAX -->
    </tbody>
</table>

@section Scripts {
    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
    <script>
        function loadStudents() {
            $.ajax({
                url: '/Student/GetStudentsJson',
                type: 'GET',
                success: function (data) {
                    const tbody = $('#studentTable tbody');
                    tbody.empty();

                    data.forEach(function (student) {
                        tbody.append(`<tr>
                            <td>${student.name}</td>
                            <td>${student.email}</td>
                            <td>${student.phone}</td>
                            <td>${student.subscribed}</td>
                        </tr>`);
                    });
                },
                error: function () {
                    alert('Error fetching data');
                }
            });
        }

        $(document).ready(function () {
```

```
        loadStudents();
    });
  </script>
}
```

Add View with AJAX (Add.cshtml)

```
@{
  ViewData["Title"] = "Add Student";
}

<h2>Add Student (AJAX)</h2>
<form id="addStudentForm">
  <input type="text" id="Name" placeholder="Name" required />
  <input type="email" id="Email" placeholder="Email" required />
  <input type="text" id="Phone" placeholder="Phone" required />
  <label><input type="checkbox" id="Subscribed" /> Subscribed</label>
  <button type="submit">Add Student</button>
</form>

@section Scripts {
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
  <script>
    $(document).ready(function () {
      $('#addStudentForm').submit(function (e) {
        e.preventDefault();

        const student = {
          Name: $('#Name').val(),
          Email: $('#Email').val(),
          Phone: $('#Phone').val(),
          Subscribed: $('#Subscribed').is(":checked")
        };

        $.ajax({
          url: '/Student/AddStudentAjax',
          type: 'POST',
          contentType: 'application/json',
          data: JSON.stringify(student),
          success: function (response) {
            alert(response.message);
            $('#addStudentForm')[0].reset();
          },
```

```
                error: function () {
                    alert('Error adding student');
                }
            });
        });
    });
</script>
}
```

TASKS:

1. **Add `EditStudentAjax` and `DeleteStudentAjax` controller methods to the provided working code**

2. Create a basic patient management system in ASP.NET MVC without AJAX.
- **Create a Patient Model** with fields: `Id`, `Name`, `Age`, `Gender`, `Phone`, `Address`.
- **Build a Registration Form** to add patients, save data to the database.
- **Display Patients List** in a table, retrieving from the database.
- **Edit Patient**: Implement form to edit patient details and update in the database.
- **Delete Patient**: Add a "Delete" button to remove patients from the database.

3. Implement the same patient management system with AJAX for dynamic updates.
- **Create a Patient Model** (same as Question 1).
- **Build an AJAX Registration Form** to submit patient data without reloading the page.
- **Display Patients List with AJAX**: Fetch and display the patient list dynamically.
- **Edit Patient with AJAX**: Update patient details via AJAX without page refresh.
- **Delete Patient with AJAX**: Remove patient from the database and update the list dynamically using AJAX.